

BEES AND WASPS

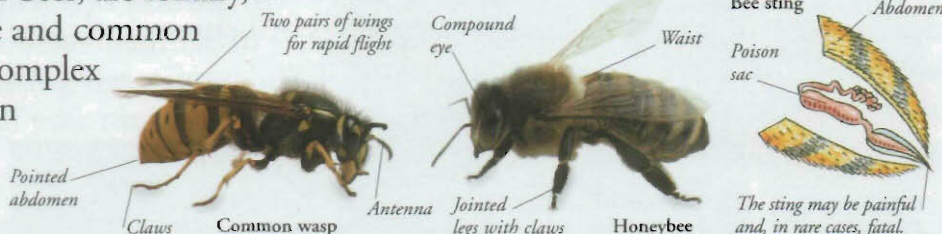
THEIR STINGS USUALLY bring these insects to our attention. However, by pollinating crops and killing pests, bees and wasps play vital roles in the world we live in. There are 115,000 species of bees and wasps. Most, such as carpenter bees, are solitary, but some, including the honeybee and common wasp, are social insects, living in complex colonies. People keep honeybees in hives for their honey and wax.

Features of bees and wasps

Bees and wasps are similar in appearance, with narrow waists between the thorax and abdomen. Most species have two pairs of wings and are excellent fliers. They have two compound eyes and three small eyes, giving them good eyesight. Bees are hairier than wasps, and are normally herbivores, while wasps are generally carnivores.

Stings

Only the females of most species of bees and wasps have stings. The sting evolved from the egg-laying tube. Wasps have unbarbed stings that they can use repeatedly for defence or to kill prey. Bees have barbed stings that cannot be extracted, causing the bee to die. Consequently, bees only sting if provoked.



Queen

Social bees and wasps have a queen in their colonies which lays eggs and runs the colony. Honeybees have one queen per hive; if two appear at the same time they fight to the death. Queens produce queen substance, a chemical that stops full sexual development of the workers.



Life cycle of a honeybee



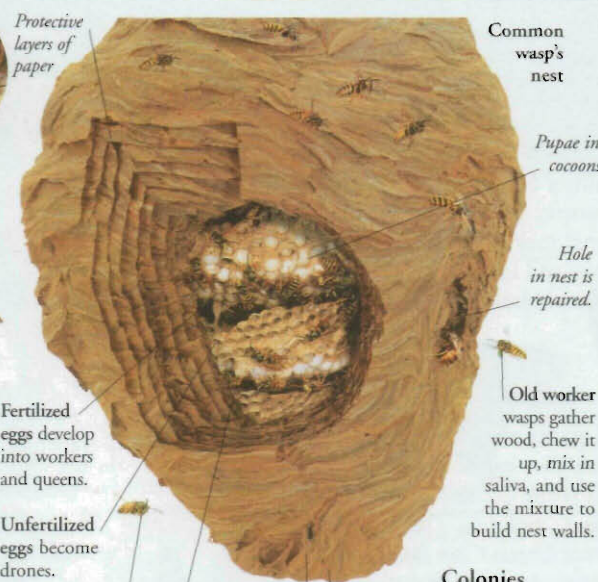
1 The queen bee spends most of her day checking cells and laying single eggs in them. She lays more than 2,000 eggs a day when there is a plentiful food supply. After 1-2 days, larvae hatch from the eggs.



2 Workers feed the larvae honey, pollen, and royal jelly. If fed extra royal jelly, larvae become queens. The larvae grow and moult, and on day 5 they spin a silk cocoon and pupate. Workers seal the cell with wax.



3 By about day 21 pupation is complete and the new adult bees have to chew their way out of the cell. Once the external skeleton has hardened and they are able to walk, the bees begin their tasks within the nest.



Nests

Common wasps' nests have the texture of paper and are made of chewed-up wood, saliva, and water. They are usually built in hollow trees or below ground. Other wasps, such as oriental stenogaster wasps, build nests out of mud. Honeybee nests are made of wax produced from glands on the bees' abdomens. Wasp and bee nests usually contain combs - layers of six-sided cells in which the young grow.

Colonies

Social wasps and bees live in large groups called colonies. Each member of the colony has a specific duty and works for the benefit of the whole nest. Wasp colonies may contain more than one million individuals; bees' nests can exceed 70,000 in number, consisting of one queen, about 69,000 workers, and 300 drones. Both queen wasps and queen bees run their nests; drones are fertile males who mate with the queen and die soon after; workers perform many tasks from building the nest to foraging for food. Drones appear before swarming time. This is when new queens leave the nest, mate with the drones, and set up nests on their own.

Food supplies

Wasps eat fruit and insects, which they also feed to their young. Adult and larval bees feed on nectar and pollen that the adults collect. Bees do a dance to tell other bees the location of the flowers, and navigate there by using the Sun.

Pollen carried on legs



Pollen, nectar, and honey

Bees carry pollen on hairs on their legs and store nectar in their stomachs. At the nest, they regurgitate the nectar. Water evaporates, concentrating the nectar to form honey. Honey and royal jelly, a high-protein substance made by workers, are also fed to the larvae.

Types of bee and wasp



Parasitic bees

These solitary bees abandon their eggs in other bees' nests. The young then destroy the original eggs and wait to be fed by the host.



Hornets

Hornets are among the largest of the social wasps. They live in large colonies and defend their nests aggressively.



Hunting wasps

These solitary wasps paralyse other insects with a sting and lay their eggs on them. The young hatch and feed on the live host.

HONEYBEE

SCIENTIFIC NAME *Apis mellifera*

ORDER Hymenoptera

FAMILY Apidae

DISTRIBUTION Worldwide

HABITAT Nests are built in hollow trees in the wild; also cultivated in hives

DIET Pollen and nectar from flowers

SIZE Length: workers: 10-15 mm (0.4-0.6 in); queen: 15-20 mm (0.6-0.8 in)

LIFESPAN Workers live for 2-3 months; queen lives for 3-5 years

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